SHRI VAISHNAV VIDHYAPEETH VISHWAVIDHYALAYA
SHRI VAISHNAV INSTITUTE OF INFORMATION TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
MST [THEORY] EXAMINATION [OFFLINE], SESSION 2021-22

Vision of the University

"To create an educational environment that engages deep intellectual, moral and spiritual stimulation, thereby nurturing leadership."

Mission of the University

"To pioneer a 'mentoring' based education system with a culture of its own, rooted in Indian ethos and in tune with contemporary times;
To impart learning through understanding, knowledge enrichment, skill development and positive attitude formation;
To encourage innovative thinking with self discipline and social responsibility."

MST EXAMINATION [OFFLINE]

TIME: 01 Hou
MAX MARKS: 2(

PROGRAM:---B.TECH.---- **BRANCH:**-----------CSE-------------- **SPECIALIZATION:**---------IBM------------
YEAR:--------II----------------- **SEMESTER:**-----------IV-------------- **SECTION:** -----------D/E----------------
COURSE-CODE:-------BTCSI411----------- **COURSE-NAME:**--------DBMS--

Q. No.	Question	Max Marks	CO Mappe
Q.1	Explain Architecture of DBMS with Neat diagram ?	4	2
Q.2	Draw an E-R diagram for an Institute's office database; maintains data about students, programs, courses, faculties, exam, marks (grades), fee etc. Mention all necessary attributes, keys, and relationships. Assume suitable data.	4	2,4
Q.3	Consider the following relational database schema consisting of the four relation schemas: passenger (pid, pname, pgender, pcity) agency (aid, aname, acity) flight (fid, fdate, time, src, dest) booking (pid, aid, fid, fdate) Answer the following questions using relational algebra queries; (1) Get the complete details of all flights to New Delhi.	4(1,1,1,1)	2,3

COURSE OUTCOME:

CO1- Understand the uses the database schema and need for normalization.
CO2- Use different types of physical implementation of database
CO3- Evaluate business information problem and find the requirements of a problem in terms of data.
C04- Design the database schema with the use of appropriate data types for storage of data in database.

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	(2) Get the details about all flights from Chennai to New Delhi. (3) Find only the flight numbers for passenger with pid 123 for flights to Chennai before 06/11/2020. (4) Find the details of all male passengers who are associated with Jet agency.		
Q.4	Define Normalization. Find out the highest normal form in given relations with proper explaination :- (1) R(ABCDEF) {$A \rightarrow BCDEF$, $BC \rightarrow ADEF$, $DEF \rightarrow ABC$} (2) R(WXYZ) {$Z \rightarrow W$, $Y \rightarrow XZ$, $XW \rightarrow Y$}	**4(2,1,1)**	**1**
Q.5	Consider the following bank database and write given SQL queries :- Customer (customer_id, customer_name, customer_street, customer_city) Loan (loan_number, branch_name, amount) Account (account_number, branch_name, balance) (1) Create customer table. (2) Insert a loan amout rs 2 lakh at vijay nagar branch for a unique value 501. (3) Find out the total sum of all account balance in the bank, branch wise. (4) Find out the account no of account whose balance between 10000 and 20000	**4(1,1,1,1)**	**3**

COURSE OUTCOME:

CO1- Understand the uses the database schema and need for normalization.
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Total No. of Questions: 05

Total No. of Printed Pages: 02

Roll No. 20100BTBDAI07223

BTCSI411

IV Semester Examination, May – June 2022

B.Tech. (CSE-BDAI, CSE-CMCI, CSE-AII)

Data Base Management System

Choice Based Credit System (CBCS)

Time: 3 Hrs.

Maximum Marks: 60

Minimum Pass Marks: 24

Note: *All questions carry equal marks, out of which part 'A' and 'B' carry 3 marks and part 'C' carries 6 marks.*
From each question, part 'A' and 'B' are compulsory and part 'C' has internal choice.
Draw neat diagram, wherever necessary.
Assume suitable data wherever necessary.

Q.1(A) Write Short notes on: **03**
(1) Data Models (2) Schema & Instance (3) Function of DBA

(B) Why Data Independence is necessary, Justify Your Answer with Proper Explanation? **03**

(C) Explain the Different components of DBMS Architecture by drawing a suitable diagram? **06**

OR

Explain the concept of Generalization and specialization in Extended ER diagram with an example; also show the different Design constraints in Extended ER diagram?

Q.2(A) Define the following: **03**
(1) Intension & Extension (2) Aggregate Functions (3) Cartesian product

(B) Why Join Operations are required in DBMS? Briefly Explain the various types of Join. **03**

(C) Explain the Fundamental operations of Relational algebra, Also explain the term Relational query Language. **06**

OR

Differentiate between free and bound variables Also explain the safe Expression in Relational calculus?

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Q.3(A) Write short Notes on following:

I. Stored Procedure & Trigger

II. Key Constraints

III. Multivalued Dependency 03

(B) Why its needed in DBMS, to convert a Relation into a well structured Relation? 03

(C) If a relation schema is in BCNF then it is in 3NF, but if a relation schema is in 3NF then it is not necessary in BCNF. Justify your answer with an Example? 06

OR

Explain the 5NF with an Example?

Q.4(A) Explain the concept of Locking. What are different modes of locking? 03

(B) Why Query Optimization is required? Explain the cost based Query optimization Technique. 03

(C) Explain the various concurrency control Techniques. 06

OR

Explain Recovery process after system failure using checkpoint?

Q.5(A) Explain the following terms with suitable example.

(1) DCL statement (2) Group by clause (3) TCL statement. 03

(B) Why can we have at most one primary or clustering index on a file but several secondary Indexes, Explain? 03

(C) What is B^+ tree? Explain Insertion and Deletion operations with Example. 06

OR

Explain hashed File Organization in Detail.

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